LAPORAN UJIAN PRAKTIKUM SEMESTER GENAP
BASIS DATA 2 MONITORING JARINGAN

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PROGRAM STUDI DIII TEKNIK INFORMATIKA

POLITEKNIK PURBAYA TEGAL

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1. Menganalisis kebutuhan data sistem monitoring jaringan.

a. Data Perangkat

Berisi:

1) ID perangkat
2) Nama perangkat
3) Jenis perangkat
4) IP address
5) Lokasi
6) Status
7) Merek

b. Data Teknisi

Berisi:

1) ID teknisi
2) Nama teknisi
3) Nomor HP
4) Email

c. Data Monitoring

Berisi:

1) ID monitoring
2) ID perangkat
3) Tanggal monitoring
4) Status koneksi

d. Data Aktivitas

Berisi:

1) ID aktivitas
2) ID teknisi
3) ID perangkat
4) Tanggal aktivitas
5) Keterangan aktivitas

2. Membuat Entity Relationship Diagram (ERD)

3. Membuat database dan tabel relasional.

a. Create table Perangkat

```
MariaDB [(none)]> use monitoring_jaringan;
Database changed
MariaDB [monitoring_jaringan]> create table perangkat (
    -> id_perangkat varchar(20) primary key,
    -> nm_perangkat varchar(100),
    -> jenis_perangkat varchar(50),
    -> ip_adress varchar(20),
    -> lokasi varchar(20),
    -> status varchar(20)
    -> );
Query OK, 0 rows affected (0.024 sec)
```

b. Create table Teknisi

```
MariaDB [monitoring_jaringan]> create table teknisi (
    -> id_teknisi varchar(20) primary key,
    -> nm_teknisi varchar(100),
    -> nohp_teknisi varchar(15),
    -> email_teknisi varchar(100)
    -> );
Query OK, 0 rows affected (0.020 sec)
```

c. Create table Aktivitas

```
MariaDB [monitoring_jaringan]> create table aktivitas (
    -> id_aktivitas varchar(20) primary key,
    -> id_teknisi varchar(20),
    -> id_perangkat varchar(20),
    -> tanggal_aktivitas date,
    -> keterangan text
    -> );
Query OK, 0 rows affected (0.023 sec)
```

d. Create table Monitoring

```
MariaDB [monitoring_jaringan]> create table monitoring (
    -> id_monitoring varchar(20) primary key,
    -> id_perangkat varchar(20),
    -> tgl_monitoring date,
    -> status_koneksi varchar(20)
    -> );
Query OK, 0 rows affected (0.024 sec)
```

```
MariaDB [monitoring_jaringan]> show tables;
+-------------------------------+
| Tables_in_monitoring_jaringan |
+-------------------------------+
| aktivitas                     |
| monitoring                    |
| perangkat                     |
| teknisi                       |
+-------------------------------+
4 rows in set (0.002 sec)

MariaDB [monitoring_jaringan]> desc aktivitas;
+-------------------+-------------+------+-----+---------+-------+
| Field             | Type        | Null | Key | Default | Extra |
+-------------------+-------------+------+-----+---------+-------+
| id_aktivitas      | varchar(20) | NO   | PRI | NULL    |       |
| id_teknisi        | varchar(20) | YES  |     | NULL    |       |
| id_perangkat      | varchar(20) | YES  |     | NULL    |       |
| tanggal_aktivitas | date        | YES  |     | NULL    |       |
| keterangan        | text        | YES  |     | NULL    |       |
+-------------------+-------------+------+-----+---------+-------+
5 rows in set (0.017 sec)

MariaDB [monitoring_jaringan]> desc monitoring;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| id_monitoring  | varchar(20) | NO   | PRI | NULL    |       |
| id_perangkat   | varchar(20) | YES  |     | NULL    |       |
| tgl_monitoring | date        | YES  |     | NULL    |       |
| status_koneksi | varchar(20) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
4 rows in set (0.015 sec)
```

```
MariaDB [monitoring_jaringan]> desc perangkat;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| id_perangkat    | varchar(20)  | NO   | PRI | NULL    |       |
| nm_perangkat    | varchar(100) | YES  |     | NULL    |       |
| jenis_perangkat | varchar(50)  | YES  |     | NULL    |       |
| ip_adress       | varchar(20)  | YES  |     | NULL    |       |
| lokasi          | varchar(20)  | YES  |     | NULL    |       |
| status          | varchar(20)  | YES  |     | NULL    |       |
| merek           | varchar(50)  | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
7 rows in set (0.010 sec)

MariaDB [monitoring_jaringan]> desc teknisi;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| id_teknisi    | varchar(20)  | NO   | PRI | NULL    |       |
| nm_teknisi    | varchar(100) | YES  |     | NULL    |       |
| nohp_teknisi  | varchar(15)  | YES  |     | NULL    |       |
| email_teknisi | varchar(100) | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
4 rows in set (0.015 sec)
```

Relasi antar tabel pada database monitoring jaringan yaitu:

- Tabel monitoring terhubung dengan tabel perangkat melalui id_perangkat
- Tabel aktivitas terhubung dengan tabel teknisi melalui id_teknisi
- Tabel aktivitas terhubung dengan tabel perangkat melalui id_perangkat

4. Mengimplementasikan query DDL dan DML

A. Implementasi query DDL

1) Membuat database

```
MariaDB [(none)]> create database monitoring_jaringan;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> use monitoring_jaringan;
Database changed
```

2) Menggunakan database

```
MariaDB [(none)]> create database monitoring_jaringan;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> use monitoring_jaringan;
Database changed
```

3) Membuat table

```
MariaDB [(none)]> use monitoring_jaringan;
Database changed
MariaDB [monitoring_jaringan]> create table perangkat (
    -> id_perangkat varchar(20) primary key,
    -> nm_perangkat varchar(100),
    -> jenis_perangkat varchar(50),
    -> ip_adress varchar(20),
    -> lokasi varchar(20),
    -> status varchar(20)
    -> );
Query OK, 0 rows affected (0.024 sec)
```

4) Mengubah Struktur Tabel

```
MariaDB [monitoring_jaringan]> desc perangkat;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| id_perangkat    | varchar(20)  | NO   | PRI | NULL    |       |
| nm_perangkat    | varchar(100) | YES  |     | NULL    |       |
| jenis_perangkat | varchar(50)  | YES  |     | NULL    |       |
| ip_adress       | varchar(20)  | YES  |     | NULL    |       |
| lokasi          | varchar(20)  | YES  |     | NULL    |       |
| status          | varchar(20)  | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
6 rows in set (0.016 sec)

MariaDB [monitoring_jaringan]> alter table perangkat
    -> add merek varchar(50);
Query OK, 0 rows affected (0.019 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [monitoring_jaringan]> desc perangkat;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| id_perangkat    | varchar(20)  | NO   | PRI | NULL    |       |
| nm_perangkat    | varchar(100) | YES  |     | NULL    |       |
| jenis_perangkat | varchar(50)  | YES  |     | NULL    |       |
| ip_adress       | varchar(20)  | YES  |     | NULL    |       |
| lokasi          | varchar(20)  | YES  |     | NULL    |       |
| status          | varchar(20)  | YES  |     | NULL    |       |
| merek           | varchar(50)  | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
7 rows in set (0.016 sec)
```

B. Implementasi query DML

1) Menambahkan Data Perangkat

```
MariaDB [monitoring_jaringan]> insert into monitoring values
    -> ('M001', 'P001', '2026-05-21', 'Online'),
    -> ('M002', 'P002', '2026-05-21', 'Offline'),
    -> ('M003', 'P003', '2026-05-21', 'Online');
Query OK, 3 rows affected (0.014 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

2) Menampilkan Data Perangkat

```
MariaDB [monitoring_jaringan]> select*from aktivitas;
+--------------+------------+--------------+-------------------+-----------------------------+
| id_aktivitas | id_teknisi | id_perangkat | tanggal_aktivitas | keterangan                  |
+--------------+------------+--------------+-------------------+-----------------------------+
| A001         | T001       | P001         | 2026-05-21        | Pengecekan router utama     |
| A002         | T002       | P002         | 2026-05-21        | Perbaikan switch lantai 1   |
| A003         | T001       | P003         | 2026-05-21        | Restart access point        |
+--------------+------------+--------------+-------------------+-----------------------------+
3 rows in set (0.001 sec)

MariaDB [monitoring_jaringan]> select*from teknisi;
+------------+--------------+-------------+----------------+
| id_teknisi | nm_teknisi   | nohp_teknisi | email_teknisi |
+------------+--------------+-------------+----------------+
| T001       | Budi Santoso | 08123456789 | budi@gmail.com |
| T002       | Andi Wijaya  | 08234567890 | andi@gmail.com |
+------------+--------------+-------------+----------------+
2 rows in set (0.001 sec)

MariaDB [monitoring_jaringan]> select*from monitoring;
+---------------+--------------+----------------+----------------+
| id_monitoring | id_perangkat | tgl_monitoring | status_koneksi |
+---------------+--------------+----------------+----------------+
| M001          | P001         | 2026-05-21     | Online         |
| M002          | P002         | 2026-05-21     | Offline        |
| M003          | P003         | 2026-05-21     | Online         |
+---------------+--------------+----------------+----------------+
3 rows in set (0.001 sec)
```

3) Mengubah Data Perangkat

```
MariaDB [monitoring_jaringan]> update perangkat
    -> set status='Offline'
    -> where id_perangkat='P001';
Query OK, 1 row affected (0.013 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```